

Improving Readiness of Communities for Evacuations with Electric Vehicles

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Abstract: This paper identifies key challenges that arise when electric vehicles (EVs) participate in evacuations, including limited charging availability and the vulnerability of the electrical grid to outages. We propose a set of recommendations to improve community readiness for EV-based evacuations, including strategic pre-positioning of mobile charging assets, EV-friendly evacuation routing and stochastic methods to estimate emergency charging demand. The findings help communities and emergency planners to increase their state of readiness for EV-based evacuations caused by natural disasters.

Introduction

California is rapidly embracing electric vehicles (EVs) as a means to decarbonize its transportation sector. According to Executive Orders N-27-25 and N-79-20, vehicles with zero (local) emissions will be the only new vehicles allowed to be sold in the state by 2035. At the same time, California is highly vulnerable to severe weather events and hazards, such as wildfires, floods, and earthquakes, which pose a significant threat to human life and property. In the past decade, all 58 California counties have experienced multiple disasters, with wildfires being the most common hazard [1]. It is expected that the frequency and intensity of weather events will continue to increase due to the impact of climate change.

State and local officials often request populations to evacuate from danger areas in order to reduce loss of human life. More than 1 million people were ordered to evacuate in California between 2017-2019 due to wildfires, often with little advance notice [2]. The evacuation strategies that our communities have relied on up to now depend on vehicles powered by fossil fuels. These vehicles can be quickly refueled by an abundant network of gas stations, while their long drive range makes them particularly valuable for short-notice evacuations. As we transition toward cleaner electric vehicles, which do not (yet) have a dense and robust re-charging network, several concerns emerge about the preparedness of our communities to effectively support short-notice evacuations using EVs.

Figure 1 provides an overview of a typical evacuation scenario in which a natural hazard (e.g., wildfire) triggers the evacuation of a community. Evacuees travel in their personal vehicles from the impacted area to a shelter, which is usually recommended by local authorities, where they can remain until conditions are safe to return home. Ideally, evacuees have sufficient energy in their EVs to reach these shelters without needing to recharge and can travel without encountering significant traffic congestion. In practice, however, these assumptions often do not hold, leading to complications during the evacuation as depicted in Figure 2.

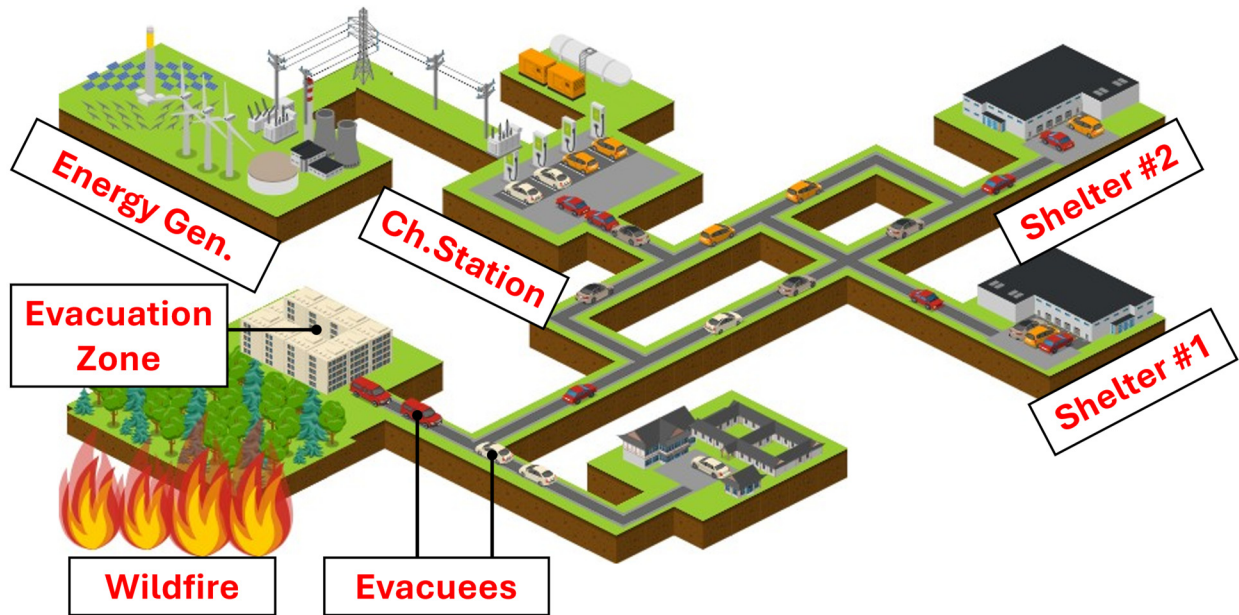


Figure 1. Example of a potential evacuation of a community caused by wildfire.

Availability of charging infrastructure and re-charging times are some of the complications, not only during day-to-day operation of EVs, but more importantly during evacuations. For example, battery-based EVs might take a long time to recharge, especially if only slow Level 1 or Level 2 charging is available (up to 0.5 miles/minute), which can be problematic during emergencies, where time is of the essence. Quicker charging is possible using DC charging (up to 10 miles/minute) [3], but the availability of this infrastructure is still limited. For example, according to the EV dashboard of California Energy Commission [4], two counties in California (Alpine, Sierra) have no public DC fast charging as of November 2025, which might delay EV evacuations in these areas. Additionally, charging infrastructure is unevenly distributed, leaving some communities underserved [5], which is more critical than mere inconvenience during evacuation.

Power outages in charging stations are another key vulnerability. This loss of power might happen due to natural disasters, e.g., physical damage to the electrical generation, transmission and/or distribution network. It might also occur due to public safety power shutoffs, which are used by electrical utilities to reduce the risk of ignition and thus wildfires; this interruption of electricity can last for several days. What if EV owners are asked to evacuate quickly and need to recharge their vehicles during a blackout

and/or public safety power shutoff? If the EV charging stations (private or public) do not have backup power, these users may be stranded, compromising their safety.

Traffic congestion is another concern, as a large number of vehicles (both electric and non-electric) may need to access the road network simultaneously to reach safety. This was evident during the 2021 Caldor Fire in California, where evacuees reported spending hours stuck in traffic jams [6]. This issue becomes even more critical when the evacuation window is short and there are a limited number of available routes to reach safety. This is especially true for rural areas. Rural areas are more vulnerable to natural disasters with limited road capacities and fewer routes available.

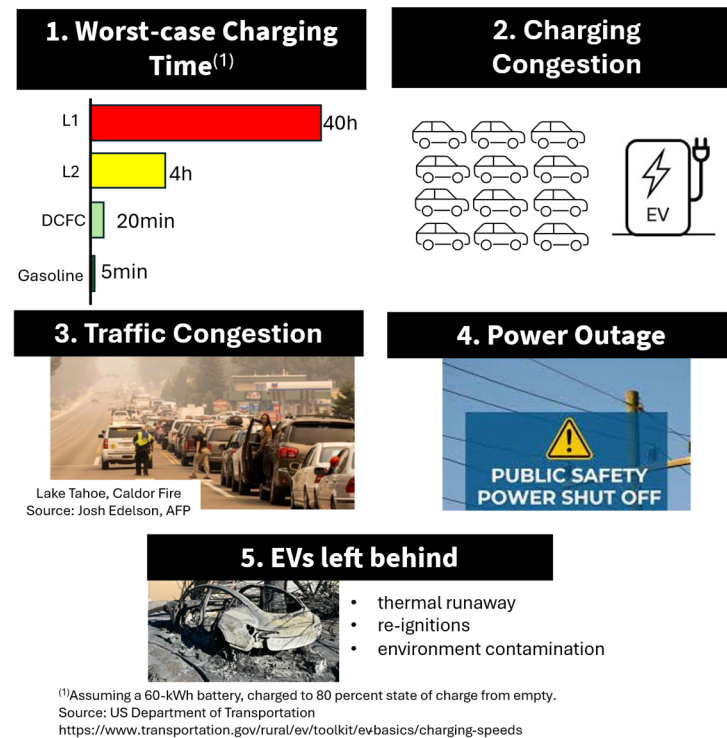


Figure 2. Challenges that may affect emergency evacuations of communities with EVs.

Electric vehicles that are left behind during the evacuation also pose additional risks. When their lithium-ion batteries ignite, the resulting fires can be significantly more intense and release toxic substances that complicate the cleanup and recovery efforts. There is an urgent need to revisit evacuation strategies of communities, which were originally designed around vehicles with internal combustion engines and may be unsuitable to cope with large-scale adoption of EVs.

The objective of this paper is to provide an overview of practical recommendations that communities can adopt to improve their preparation for emergency evacuations using EVs. These recommendations are based on the research that the authors have been developing as part of the California Climate Action Initiative [17]. They are also based on interviews and public workshops with numerous entities in

California, including emergency planners, industry, non-profit organizations and electrical utilities, which are summarized in the last part of the manuscript.

Recommendations for EV Evacuations

This section provides an overview of strategies to improve preparedness for EV evacuations. They are divided into three main areas: i) charging infrastructure; ii) transportation; iii) communication and partnerships. As a use case for this discussion, we consider the Mariposa county in California, USA, a region that frequently experiences wildfire-induced evacuations, such as the 2017 Detwiler Fire, 2018 Ferguson Fire, and the 2022 Oak Fire (see Figure 3). From a transportation perspective, this region has a small number of roads that can be used to escape a fire, which poses additional risks for evacuations.

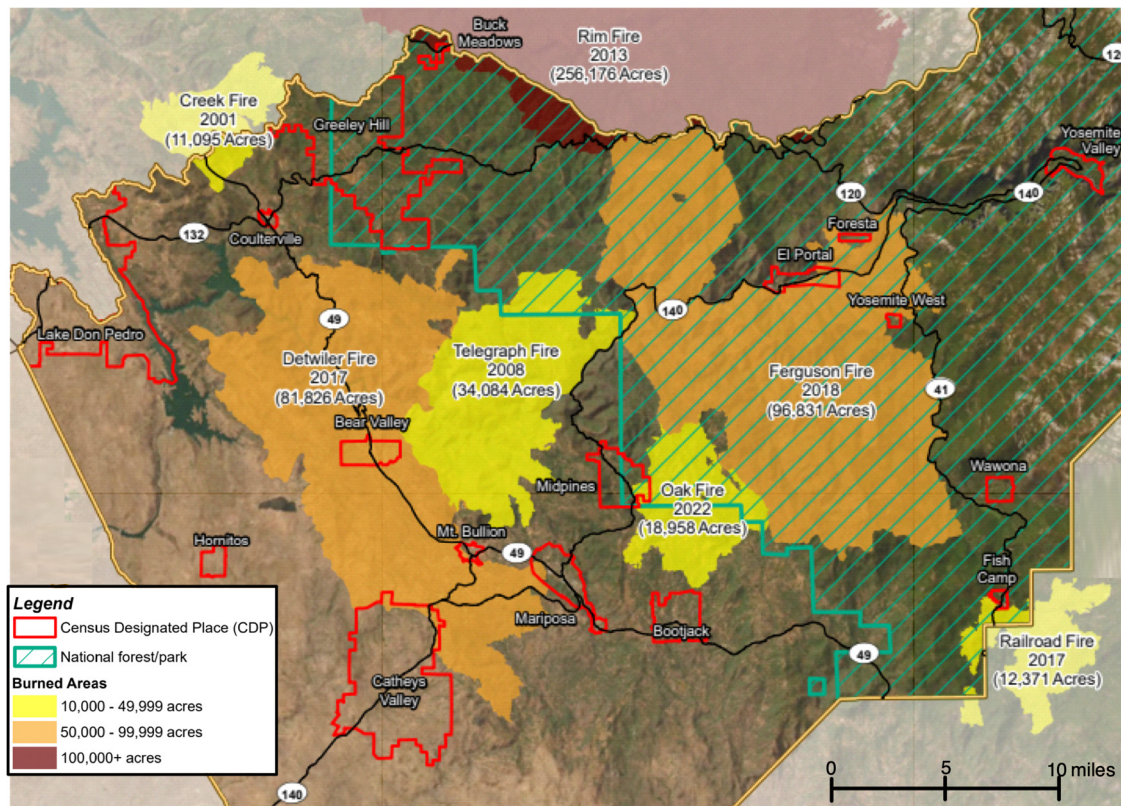


Figure 3. Overview of the Mariposa county, California (USA), including main roads (HW 140, 41, 49, 132, 120) and recent wildfires. (Adapted from Mariposa's Local Hazard Mitigation Plan).

I) Charging Infrastructure

#1 Communities need to identify potential gaps in charging energy and power that may emerge during EV evacuations

During EV-based evacuations, there may be a surge of energy needs at charging stations. This charging energy depends on several parameters (θ_{evac}), including: i) distance between evacuation zone and

shelters, ii) number of vehicles used in the evacuation, iii) initial SoC of the vehicles; and iv) energy efficiency of the vehicle.

To assess the evacuation charging needs, stochastic methodologies [10] can be used. This approach models evacuation parameters (θ_{evac}) as random variables and employs Monte Carlo simulation techniques to quantify charging energy requirements. This methodology is computationally efficient, easily accessible for local officials and captures the uncertainties in many of these parameters.

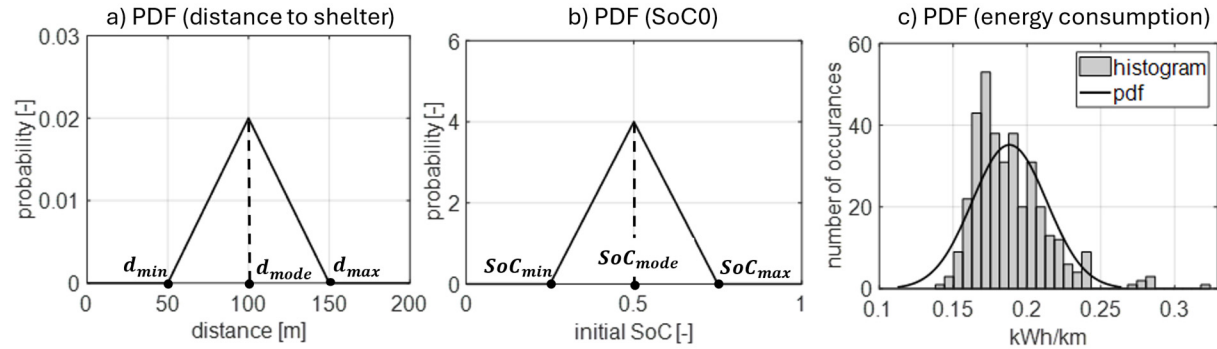


Figure 4. Plots a) and b) depict the triangular probability distribution functions (PDF) for the distance to shelter and initial state of charge (SoC); the points $w_d = [d_{min}, d_{mode}, d_{max}]$, $w_{SoC} = [SoC_{min}, SoC_{mode}, SoC_{max}]$ represent parameters that users can adjust based on the evacuation scenario. Plot c) represents the histogram of the most common EVs on the market (source: www.evdatabase.org) and the PDF fitted with a Gaussian distribution.

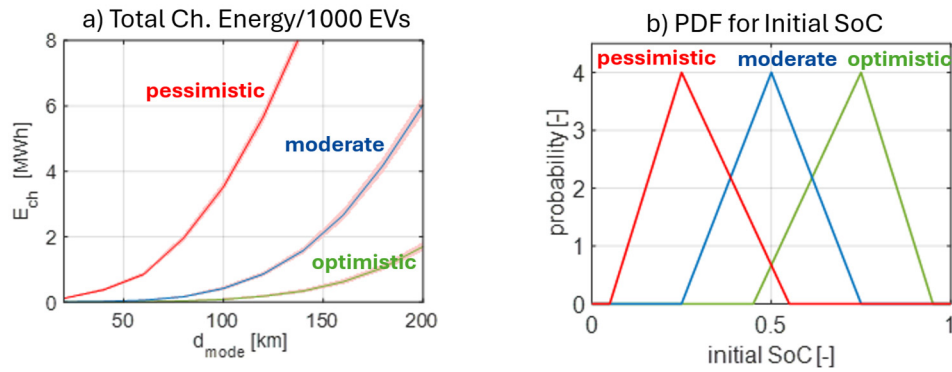


Figure 5. Plot a) depicts the sensitivity analysis of EV charging energy (normalized per 1,000 vehicles) as a function of the (i) most likely distance to the shelter (d_{mode}) and (ii) three different scenarios for the triangular PDF of the initial SoC (pessimistic, moderate, and optimistic), which are illustrated in plot (b).

Figure 4 provide an example of the stochastic distribution of some of the key evacuation parameters θ_{evac} . In this analysis, we assumed a triangular probability distribution function (PDF) for the evacuation distance because it provides finite support; and it can also be parameterized by emergency planners and community stakeholders using intuitive quantities, such as minimum (d_{min}), maximum (d_{max}), and most-likely (d_{mode}) evacuation distance. Similarly, we also assumed a triangular distribution for the initial SoC (Figure 4b), which provides an easy-to-parameterize PDF with minimum, maximum and most-likely initial SoC (SoC_{min} , SoC_{max} , SoC_{mode}).

For the energy consumption of the EVs (Figure 4c), we assumed a Gaussian distribution, which was parameterized to approximate the energy consumption of the most common EVs on the markets, reported by the public database www.evdatabase.org.

Figure 5a represents the average charging energy per 1000 vehicles as a function of the (most common) evacuation distance¹ (d_{mode}) under three scenarios for the initial SoC (i) a pessimistic initial SoC, in which most evacuees have a low driving range; (ii) a moderate initial SoC; and (iii) an optimistic initial SoC, in which almost all evacuees have at least 50% SoC (see Figure 5b). As expected, the charging energy increases with the evacuation distance. The initial SoC is another crucial factor: for an evacuation distance of 100km, the charging needs may vary from $E_{ch} = 0.1\text{MWh}/1000$ vehicles (optimistic initial SoC scenario) to $E_{ch} = 3.5\text{MWh}/1000$ vehicles (pessimistic initial SoC). If grid power is not available, this charging energy will need to be transferred to the EVs using backup energy sources or mobile chargers. This wide range of charging energy poses a challenge for emergency planners: should they aim for the optimistic case of 0.1MWh/1000 vehicles of charging energy (which can be fulfilled with a small mobile charger), or should they plan for the worst-case charging needs (3.5MWh/1000 vehicles), which may require multiple backup/mobile charging units and increase planning costs? In practice, it will be difficult to anticipate the initial SoC of the evacuees. This variable, which has a profound impact on charging energy needs, represents one of the greatest sources of uncertainty in planning evacuations involving EVs.

Another factor to consider is the number of charging ports (n_{ports}) available during the evacuation, as well as their power limits. Assuming the existence of a time window for the evacuation (T_{evac}), the average charging power needed per port can be computed as

$$p_{ch}^{port} = \frac{E_{ch}}{n_{port} T_{evac}}$$

Table 1 provides an overview of the average port power for different combinations of n_{ports} and E_{ch} , while assuming $T_{evac} = 3\text{h}$ (a possible evacuation window due to a wildfire). As expected, when the charging energy is high and the number of ports is low, the average charging power goes up. In the example illustrated in Table 1, most of the scenarios require fast charging ($>20\text{kW}$), which is not yet widespread in many communities.

¹ We consider $d_{min} = \max(0, d_{mode} - 50\text{km})$ and $d_{max} = d_{mode} + 50\text{km}$.

		Ech=Charging Energy per 1000 vehicles			
		1000 kWh	2000 kWh	3000 kWh	4000 kWh
# ports	10	33.3 kW	66.7 kW	100.0 kW	133.3 kW
	20	16.7 kW	33.3 kW	50.0 kW	66.7 kW
	30	11.1 kW	22.2 kW	33.3 kW	44.4 kW
	40	8.3 kW	16.7 kW	25.0 kW	33.3 kW

L2 Charging

Fast Charging

Table 1. Average charging power required per charging port, assuming an evacuation window of $T_{evac} = 3h$. Cells in “blue” represent an average requirement for fast charging. Columns represent amount of energy per 1000 vehicles.

Returning to the Mariposa evacuation use case: this county has 144 charging ports [4]. According to California’s Department of Motor Vehicles, there are 13,067 vehicles registered in the county². Consider the hypothetical (future) scenario where all vehicles are electric and need to be charged at the existing public stations, leading to approximately 11 charging ports per 1000 EVs. To deliver the charging energy for the moderate-to-pessimistic initial SoC scenarios discussed above, the charging ports would need to be DC fast charging (see Table 1). This contrasts with today’s reality, where only 23% of the charging ports offer DC fast charging [4].

#2 Strategic deployment of mobile EV chargers helps decrease waiting and evacuation time

Fixed charging infrastructure, which is generally optimized for routine daily use, may be insufficient to provide the charging energy and power during short-notice evacuations. Mobile charging stations (MCSs) can support fixed charging infrastructure during periods of high charging demand. These mobile resources are usually composed of energy storage units (such as batteries, fuel cells, diesel, and natural gas) that are installed on a trailer or on the back of a truck, enabling quick deployment. Two of the key questions that arise when using MCSs in evacuations include: how many MCSs should be used and where to place them.

To address these questions, optimization-based frameworks can be developed in order to guide the strategic placement of this resource [7]. Because the positioning of MCS affects evacuation routing, these frameworks typically integrate path selection with deployment locations to minimize the aggregate driving and waiting time [8].

To better understand the potential of MCS, we applied the MCS placing/routing methodology from [7] to two EV population scenarios within the context of the Mariposa evacuation case study. The first

² The number of vehicles to evacuate could be significantly higher, since Mariposa County has many visitors/tourists that increase the town’s population, especially during the summer.

scenario assumes a homogeneous EV population, where all EVs share identical battery capacities (30kWh) and initial state of charge (=20%). The second scenario considers heterogeneous battery capacities, with battery sizes drawn from a discrete uniform distribution over the set [20kWh, 30kWh, 40kWh], while the initial state of charge (SoC) was sampled from a truncated Gaussian distribution with mean, variance, truncated lower and upper bound equal to 20%, 10%, 0%, and 100% respectively. As shown in Figure 6a, by adding two MCSs, Mariposa can decrease the average evacuation time by 14% for the homogeneous case and 16% for the heterogeneous case, while the maximum (clearance) time³ fell by almost 25%. With four MCSs, we can observe reductions of 20% in average evacuation and 30% in maximum time. These improvements are due to shorter vehicle waiting times at the EV chargers, which were directly influenced by the availability of a larger number of charging ports. Figure 6b further shows that adding a couple of MCSs (each assumed to provide five additional charging ports) reduced average wait time by 25%, while four MCSs achieved reductions of 35–44%. Maximum wait times dropped up to 32–45%. These benefits, especially pronounced in heterogeneous fleets, stemmed from the ability of MCSs to not only add charging capacity but also enable new routing options and more flexible departure schedules, improving the utilization of existing fixed charging stations [7] [9].

Despite these potential benefits in reducing evacuation and waiting times, it is important to acknowledge that the deployment of MCSs will be ultimately constrained by economic considerations (e.g., how many MCS units a community or region can afford to acquire) as well as logistical issues (e.g., how to recharge or refuel the MCSs during emergency conditions, particularly when the charging demand exceeds the onboard energy-storage capacity of the MCSs). One attractive option to deal with economic costs consists of exploiting joint ownership, e.g. state government and/or utilities maintain mobile charging fleets that can be quickly dispatched to high-risk regions, reducing the burden on smaller jurisdictions [13]. Another possible option is renting MCSs during periods when the likelihood of hazard occurrence is high, which could also reduce the communities' upfront investment costs.

³ Time necessary to bring all evacuees to safety.

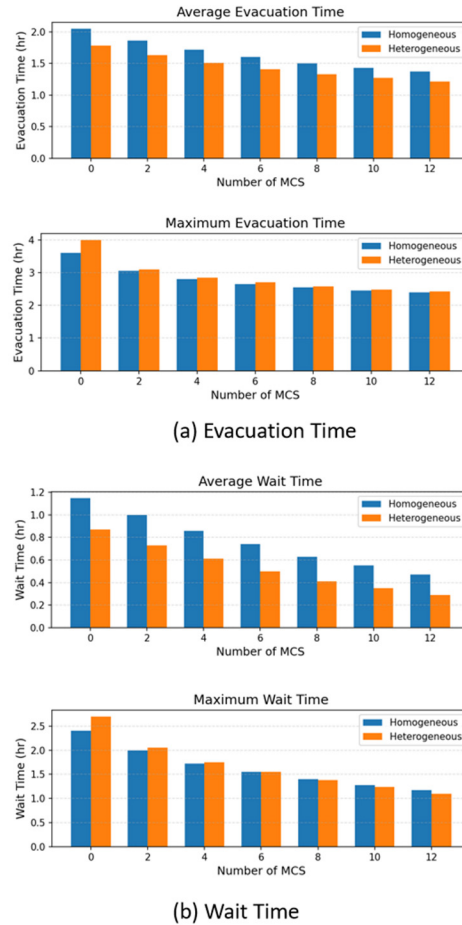


Figure 6. Improvement in a) evacuation time and b) waiting time at the charging stations as a function of the number of mobile chargers deployed in Mariposa.

#3 Fixed Charging Stations are vulnerable to grid outages and may necessitate integration of backup power solutions or mobile charging support

Power outages at charging stations represent a critical vulnerability during EV evacuations, particularly under wildfire-induced transmission damage or Public Safety Power Shutoffs (PSPS). Without backup power, evacuees may be stranded mid-evacuation, directly compromising public safety. Fixed Charging Stations (FCSs) connected to the transmission network are especially exposed, as their operation depends on both grid availability and real-time power flow conditions.

Our interactions with community partners (summarized at the end of this manuscript) emphasized that many counties rely on a limited number of transmission corridors, making them highly susceptible to disruption during wildfires. Even a single line outage (an N-1 event) can trigger significant power flow redistribution, potentially isolating entire charging regions when demand is highest. To assess charging reliability under such conditions, Optimal Power Flow (OPF) models with N-1 contingency criteria can be employed [16]. Within this framework, charging demand at FCSs may be partially or fully curtailed to alleviate line congestion, not only due to physical charger failure, but also as a corrective operational response to maintain system stability.

Figure 7 provides an illustrative benchmark network (based on the IEEE RTS-24 test system⁴) to demonstrate how OPF and N-1 contingency analysis can be applied to evaluate the resilience of FCS to disruptions in power transmission. It can be seen that the outage of a single transmission line can render the FCS at a major network hub completely unavailable (0% served), while other locations experience partial service reductions. Placing MCS at these locations is desirable to enhance resilience under unplanned grid disruptions and localized transmission failures.

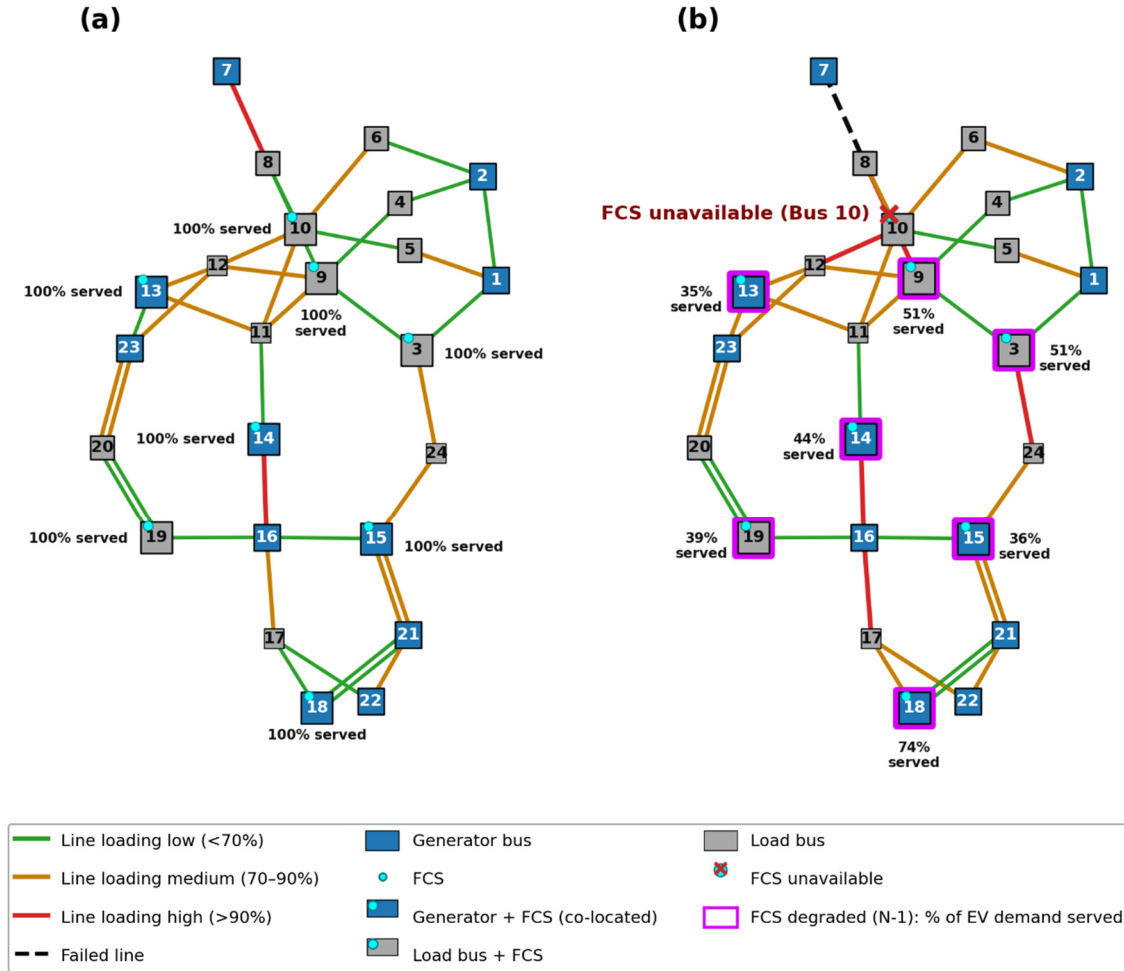


Figure 7. OPF results for the IEEE RTS-24 system under (a) normal operation and (b) N-1 contingency (line 7-8 outage). The outage redistributes power flows, increasing line loading and forcing EV demand curtailment at all FCS locations, with Bus 10's FCS becoming fully unavailable (0% served).

II) Transportation

⁴ The authors were unable to obtain the configuration and operational parameters of the actual transmission network in Mariposa County, which serves as the primary use case in this manuscript. Therefore, the widely used IEEE RTS-24 benchmark system was employed to illustrate potential vulnerabilities and operational challenges that may affect power transmission to FCSs during emergency conditions.

#4 Communities need customized evacuation routes that intelligently allocate EVs to road segments and charging stations, preventing traffic and charging bottlenecks

Emergency planners also need to provide suitable evacuation routes to the communities, which may include stopping for battery recharge. In practice, evacuees may rely more on commercial navigation platforms, which may create pitfalls for the evacuation. For example, when thousands of evacuees receive identical routes and charging suggestions, the resulting convergence of vehicles on a few major corridors can cause severe traffic congestion and even gridlock, delaying evacuation and increasing safety risks. We advocate that, in some scenarios, it is beneficial to recommend distinct paths to different evacuees based on real-time conditions and individual EV characteristics (SoC, battery capacity, driving distance) and EV charging stations (charger availability and potentially number of waiting vehicles). By routing traffic intelligently and accounting for heterogeneous EV needs, these customized routes direct drivers toward safe and energy-feasible evacuation paths while balancing demand across the network.

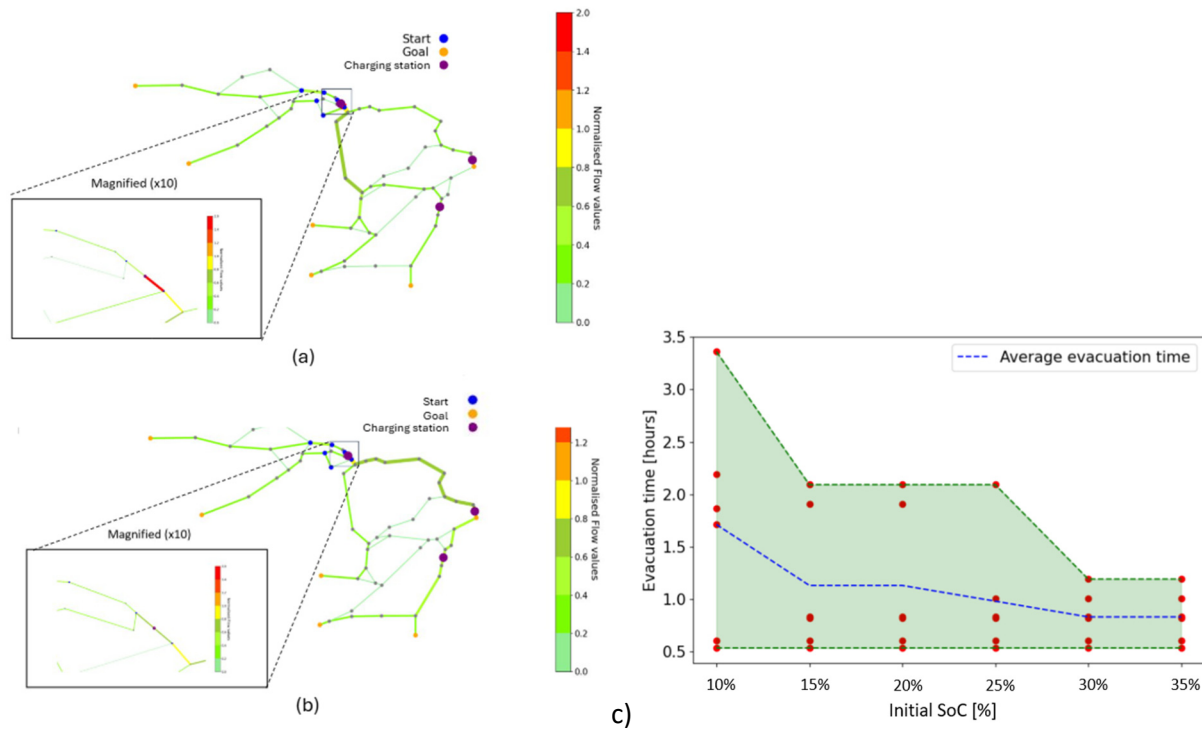


Figure 8. Comparison of the normalized vehicle concentration in Mariposa for a) uncoordinated evacuation and b) “EV-friendly” evacuation plan, which assigns different routes to evacuee groups. The figure c) shows the average evacuation time for the Mariposa evacuation use case, as function of initial SoC.

Figure 8a illustrates the uncoordinated emergency evacuation scenario for the Mariposa use case, where the majority of the evacuees follow the same route and charging pattern. The color of each road segment represents the normalized *vehicle concentration* of that road relative to maximum free-flow conditions (i.e., uncongested traffic). Therefore, green indicates that the vehicle concentration is less than the maximum free flow concentration and red indicates that the vehicle concentration is greater than maximum free flow concentration and will cause traffic congestion in that road segment. This

uncoordinated evacuation results in a large concentration of vehicles converging in a few road links, leading to significant traffic buildup and delays.

In contrast, Figure 8b presents an EV-aware coordinated evacuation strategy designed to alleviate such congestion. Here, EVs are intelligently assigned to different routes and charging stations across the network based on their individual SoC levels and charging requirements. The coordinated evacuation strategy is generated through an integrated mixed integer linear programming (MILP) optimization framework that jointly determines routing and charging [12]. Rather than sending all EVs along the shortest path, the MILP allocates traffic across multiple corridors, and selects distinct charging options, even when evacuees take the same physical route, to smooth both spatial and temporal demand at charging nodes. This coordinated allocation effectively distributes both the traffic and charging demand, thereby preventing local bottlenecks and improving overall evacuation efficiency.

It is important to note that this EV-aware coordinated evacuation strategy assumes that all evacuees follow the assigned routes and plans. In practice, evacuees may not be aware of the route (due to lack of communication) or even refuse to follow the recommended routes due to panic or a belief that they know a better evacuation route.

III) Communication and Partnerships

#5 Educating the population to maintain high charge levels in EVs is crucial

One of the most cost-effective approaches to increase emergency preparedness consists of raising awareness among communities for the importance of keeping high SoC in their EVs when the occurrence of hazardous events is high. Research [12] [9] has shown that EVs that have high state of charge can reach most shelters without needing to stop or access charging infrastructure. Figure 8c summarizes the distribution of evacuation times across varying initial SoC for the Mariposa's evacuation use case, revealing a clear inverse relationship between initial SoC and evacuation duration. The additional time spent during evacuation arises primarily from three factors: (i) detours to reach charging stations, (ii) waiting in the charging queue, and (iii) time spent charging. The communities with high adoption of EVs are encouraged to implement communication campaigns that reinforce the need to maintain high SoC in their EVs during periods (e.g., wildfire season) when evacuation is likely.

Note that this communication strategy also has some limitations. For example, 100% public compliance is not always guaranteed; furthermore, the message may not reach the entire population, and some recipients may simply choose not to follow it or may forget to do so. There are also practical constraints that may prevent certain evacuees from maintaining a high SoC, such as a lack of access to home charging.

#6 Real-time communication between emergency planners and evacuees is crucial

Mobile devices, like smartphones, could play an important role in this dissemination of evacuation guidance for EV owners. In the future, we anticipate the need for new types of applications that can

connect emergency planners and the population. Local authorities can interactively parameterize the evacuation problem by selecting areas for evacuation and defining the number of mobile EV chargers available in the area, as well as their capacity and location (Figure 9). The population can also visualize evacuation routes and charging stations (fixed or mobile) which they can use during evacuations.

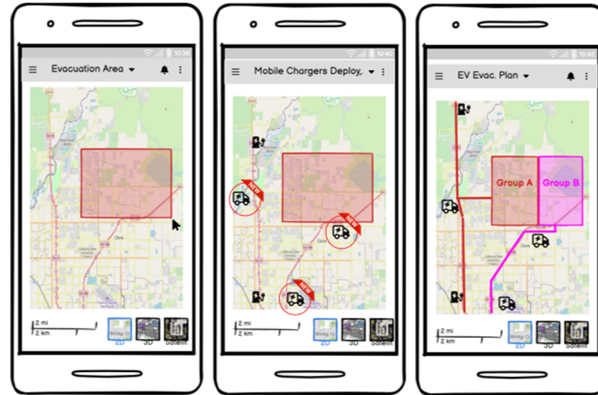


Figure 9. Mock-up of a future mobile interface that would allow different groups of evacuees to receive the recommended EV-friendly routes and (mobile and fixed) charging options.

#7 Planning for EV evacuations should integrate community perspectives to ensure the resulting strategies are appropriate and feasible

Evacuation plans developed in a “vacuum” might be theoretically sound but practically impossible to execute. Feedback from community partners who will lead and execute the plans ensures that the strategies are realistic, and reflect resources available within the community. To address this dimension, we held three workshops (4.26.2024, 12.5.2024, 10.2.2025) with numerous community partners in order to discuss recommendations in EV evacuations and also collect their feedback. This group of partners was composed of emergency planners from multiple counties in Central California, public officials from energy and transportation state agencies, representatives from electric utilities, manufacturers of charging stations, and non-profit organizations that promote the advancement of clean transportation. Figure 10 and the next paragraphs provide an overview of the key concerns from these community partners, which could stimulate further research on EV-based evacuations.

Fixed charging infrastructure: reliability of EV charging infrastructure is one of the key concerns. High failure rates were repeatedly mentioned as a risk factor; a recent study found that more than 25% of fast chargers in the Bay Area, CA, are nonfunctioning [15]. In the planning of charging needs during evacuations, emergency planners should include a safety buffer to account for malfunctioning EV chargers. Another factor that was pointed out during the workshops was the lack of interoperability between charging standards — especially between older Combined Charging System (CCS) chargers and newer North American Charging Standard (NACS) systems. Since people evacuate under stress, anything that reduces charging options becomes a major risk. Having access to universal charging adapters could help deal with this interoperability issue. Participants also recommended selecting shelters with EV charging stations so evacuees could charge their vehicles before going home. They also suggested including refueling options for other clean transportation, such as hydrogen.

Several counties also emphasized the risk of losing grid power due to wildfires or transmission-line outages. For example, some counties are dependent on single transmission lines and are vulnerable to wildfire-related power shutoffs. The community partners asked for more backup power options to provide the charging energy (E_{CH}) for EVs, including microgrids, off-grid chargers, cleaner generators, or multi-fuel backup systems. Some agencies also noted that charging demand during evacuations may coincide with peak cooling load in the summer, making grid stress even worse.

Mobile charging is generally seen as an attractive way to improve EV evacuation readiness, but they also described several barriers. Permitting for this infrastructure may take too long, especially for units that produce pollutant emissions. Some vendors require multi-week installation windows, which is not compatible with short-notice emergency conditions. Additionally, communities also need a memorandum of understanding (MOU) with possible site hosts (e.g., parking lots, public facilities, malls, etc.) to deploy these chargers during emergencies. Some of the key elements to be addressed in the MOU include locations, access hours, power interconnection requirements, traffic flow management, and liability provisions. San Diego Gas & Electric (SDG&E), for example, has recently deployed MCS equipped with a 500 kW lithium-ion battery pack to maintain power for critical customers during wildfire-related public safety power shutoff (PSPS) events, which can provide a template for other communities and utilities to follow [14] [19].

Emergency Routing: The community partners mentioned the importance of evacuating not only passenger EVs, but also medium- and heavy-duty EV vehicles, which need more space and consume more energy. Many participants asked for routing recommendations that explicitly avoid guiding people into areas that may experience hazardous conditions, something current traffic routing tools sometimes fail to offer. They also requested the ability to compare routes based not only on travel time but also on charging availability and energy demand. Using shared transportation solutions (e.g., buses) was also discussed as an option to improve evacuation efficiency. However, community partners reported that most evacuees prefer private vehicles for evacuations; their personal car is often one of the main financial assets they can take with them, and they are reluctant to leave it behind.

Communications: Community partners repeatedly noted that many evacuees do not follow instructions, being subject to panic and line-cutting at chargers. Some counties in CA also face language barriers and limited cell coverage (especially in rural areas), which reduces the ability of emergency planners to communicate instructions to evacuees. Several emergency planners noted that their hazard and evacuation plans were written before mass EV adoption and no longer reflect the current needs of the communities. They called for clear guidelines for incorporating EVs into emergency planning, and dedicated funding so that new requirements (e.g., possible mandatory backup power and mobile charging units) do not become unfunded mandates.

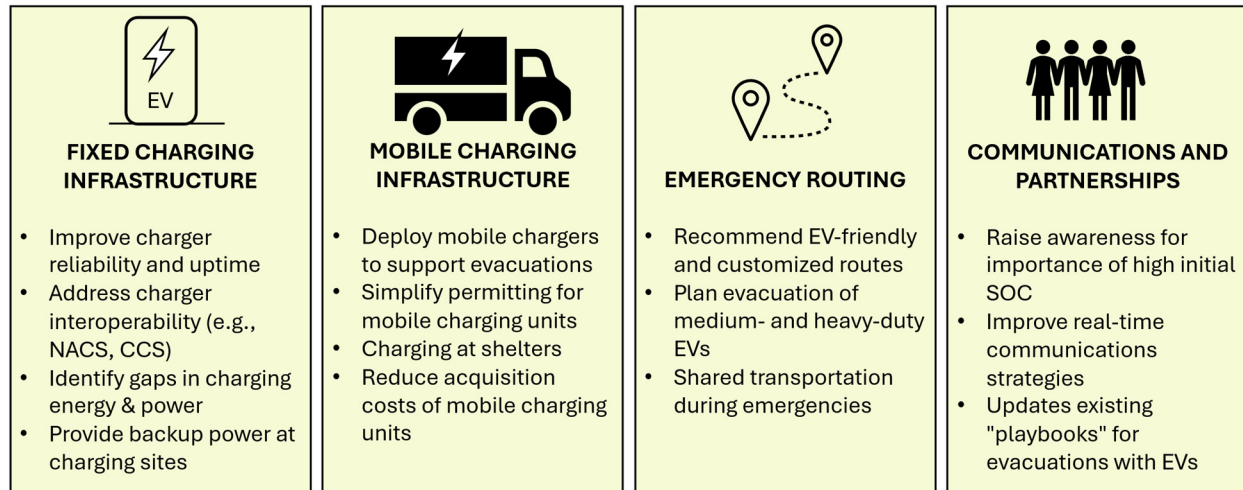


Figure 10. Overview of recommendations to improve preparedness for EV-based evacuations, jointly developed with community partners in California, including emergency planners, stage agencies, industry, and non-profit organizations.

Conclusion

As communities transition toward clean transportation, emergency evacuation planning must also evolve to address the unique constraints introduced by EVs. Our analysis summarized several practical strategies to enhance EV evacuation readiness, focusing on deployment of mobile charging infrastructure, EV-friendly traffic routing and real-time communication with potential evacuees. We also presented a summary of key issues and infrastructure gaps during EV-based evacuations, which were identified by key stakeholders, including a group of emergency planners, utilities, industry, and community organizations in Central California.

We anticipate that wealthier regions with greater technical and financial resources will be better positioned to adopt the strategies presented in this manuscript. Communities with fewer resources may require additional support from state or federal agencies to implement these strategies effectively. Additionally, vulnerable populations — such as individuals without access to home charging or those who rely on older or shorter-range EVs — may face greater challenges during evacuations and should therefore be prioritized in efforts to improve preparedness for EV-based evacuations.

Outlook

- Future work should also investigate the role of bidirectional charging technologies, including vehicle-to-grid (V2G) and vehicle-to-vehicle (V2V) systems, in improving evacuation resilience. During emergency conditions, large EVs (e.g., trucks and buses) could potentially charge other EVs by serving as mobile charging assets, while also supporting shelters and other critical infrastructure. Future research should investigate how communities could effectively leverage these assets during emergencies.

- Artificial Intelligence (AI) and machine learning could also play a role in responding to emergency evacuations. AI systems could continuously analyze evolving hazard conditions (e.g., wildfires), traffic congestion and charging-station availability and then adapt evacuation plans (e.g., recommended routes and deployment of MCS). However, because emergency evacuations are sporadic events, obtaining sufficiently large and representative datasets for training AI systems for these use cases may pose a challenge.
- Future work should integrate real power transmission and distribution network models (instead of the IEEE RTS-24 benchmark system considered in this work) together with region-specific operational constraints related to the power grid. This integration is important to enable more realistic resilience assessments (e.g., OPF under N-1 contingencies) for critical infrastructure, such as fixed charging stations, during emergency evacuations. Microgrids integrated with onsite renewable energy sources and battery energy storage systems could also play an important role in improving the resilience of EV evacuations. During grid outages, microgrids could maintain operation of critical infrastructure (e.g., charging, shelters, emergency response centers, and communication systems). These localized energy systems could reduce dependence on vulnerable transmission corridors and provide backup charging capabilities.
- Autonomous and connected vehicles present both opportunities and challenges for emergency evacuations. On the opportunity side, autonomous vehicles may be more compliant with evacuation routes recommended by authorities (potentially reducing the effects of irrational or panic-driven decisions made by humans under stressful conditions), while connectivity with other vehicles and infrastructure could improve traffic coordination and overall transportation efficiency. However, these technologies also introduce new vulnerabilities. For example, a 2025 power outage in San Francisco [18] caused significant disruptions to the fleet of autonomous vehicles operated by Waymo, with some vehicles coming to a halt in the middle of streets and intersections. Future autonomous vehicles should therefore be capable of safely navigating and operating under conditions in which transportation, communication, and charging infrastructure are significantly degraded or unavailable.
- This paper focused on pre-disaster recommendations. In the future, it would be interesting to extend this research to deal with the restoration of the clean transportation infrastructure in the aftermath of natural disasters.

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